

DATA ANALYTICS – TASK 4

DATA CLEANING & REPORTING AUTOMATION

INTRODUCTION

In data-driven environments, raw datasets often contain inconsistencies, missing values, and unstructured formats that limit their usability. This project focuses on transforming such raw marketing and sales data into a structured and analysis-ready format.

OBJECTIVE

The objective is to develop a streamlined workflow that includes data cleaning, feature engineering, and automated preprocessing using Python, followed by interactive visualization using Power BI. The goal is to identify key factors influencing revenue and evaluate marketing and customer performance.

DATA PREPARATION

The dataset consists of 60,000 records containing information related to marketing spend, customer behavior, and sales revenue.

Initial inspection revealed missing values in key columns such as customer satisfaction score, email open rate, and discount percentage. These were handled using median imputation and domain-based assumptions. The date column was converted into datetime format to enable time-based analysis.

Feature engineering was applied to create meaningful metrics such as total advertising spend, return on investment (ROI), and customer engagement. These transformations improved the analytical quality of the dataset.

	id	date	region	sales_channel	product_category	customer_segment	season	marketing_budget_usd	ad_spend_online_usd	ad_spend_offline_usd	...	customer_age
0	1	2020-11-12	Riyadh	Retail Store	Cosmetics	Regular	Q4	1664.51	952.30	326.97	...	42
1	2	2022-07-05	Dubai	Online	Cosmetics	New	Q3	2452.29	1014.25	414.76	...	24
2	3	2020-11-11	Riyadh	Online	Electronics	Regular	Q4	1026.13	328.30	242.72	...	32
3	4	2022-10-01	Cairo	Online	Electronics	Regular	Q4	1102.86	628.30	204.58	...	62
4	5	2023-12-12	Cairo	Retail Store	Food & Beverage	Corporate	Q4	2517.55	777.24	817.88	...	23

5 rows × 23 columns

DATA ANALYSIS & AUTOMATION

Exploratory analysis was conducted to understand the distribution and relationships within the dataset. Statistical summaries indicate that revenue varies widely, reflecting differences across campaigns, regions, and customer segments.

Correlation analysis shows that marketing budget and advertising spend have a strong positive influence on revenue. In contrast, customer-related metrics such as engagement and conversion rate show weaker relationships, indicating potential inefficiencies in converting user activity into sales.

ROI analysis reveals an average value of approximately 2.4, suggesting overall profitability but with significant variation across observations.

To improve efficiency, a Python-based pipeline was developed to automate data preprocessing. This pipeline integrates cleaning and feature engineering into a reusable function, ensuring consistency and scalability.

[16]: **sales_revenue_usd** **website_traffic** **marketing_budget_usd**

count	60000.000000	60000.000000	60000.000000
mean	5769.193975	22138.134183	9545.749587
std	4652.486315	40028.332078	13074.885483
min	1079.879100	241.990000	500.000000
25%	2744.180000	2961.000000	2183.567500
50%	4340.050000	8065.000000	4922.120000
75%	7091.632500	22057.000000	11040.090000
max	27666.798500	261352.820000	79109.793500

```
df['roi'].describe()
```

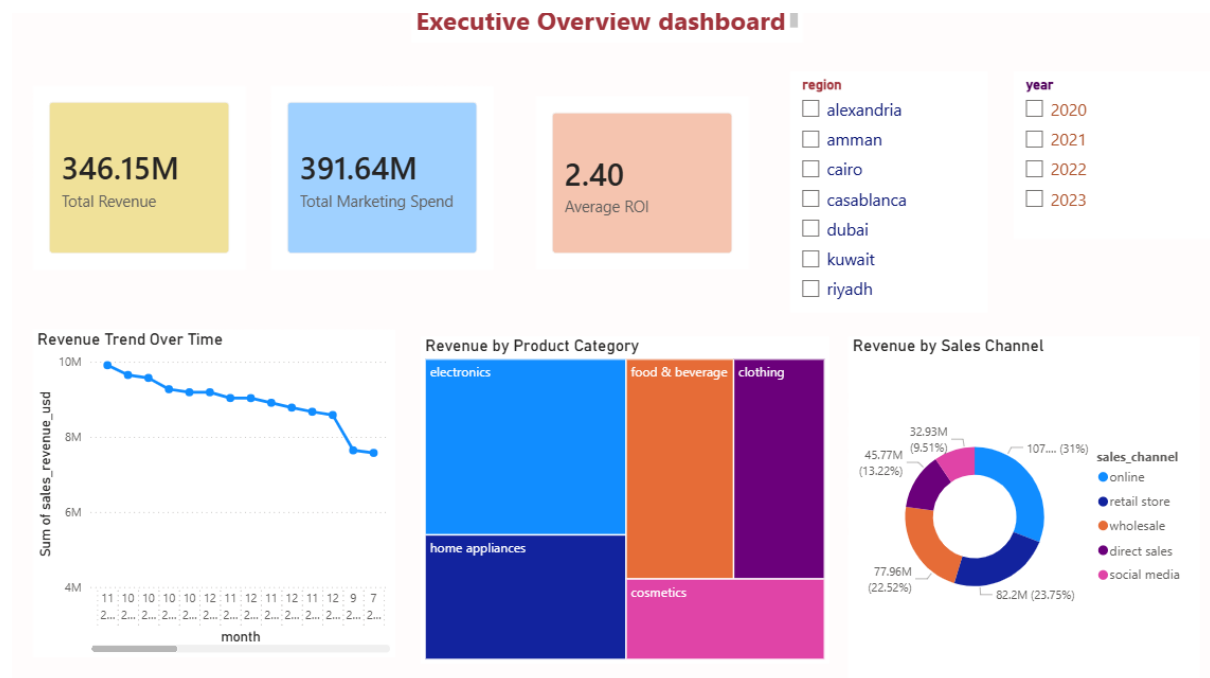
```
count    60000.000000
mean      2.404276
std       3.029617
min       0.074067
25%       0.680973
50%       1.300445
75%       2.753907
max       18.030759
Name: roi, dtype: float64
```

VISUAL ANALYSIS & INSIGHTS

1.EXECUTIVE OVERVIEW:

This section provides a high-level view of overall business performance. The total revenue generated is significant, with an average revenue of around 5,700 USD and an ROI of approximately 2.4, indicating overall profitability.

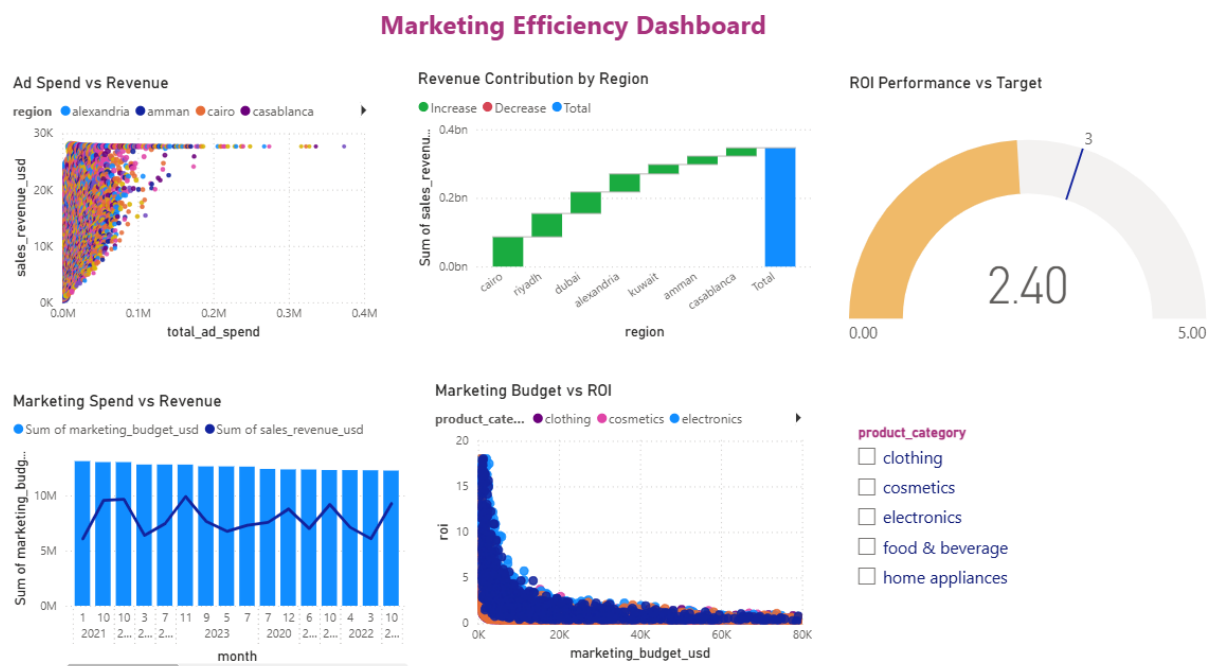
The revenue trend shows fluctuations over time, suggesting the impact of seasonal demand and marketing activities. Regional analysis highlights that certain regions contribute more significantly to revenue, indicating uneven business distribution.



2. MARKETING PERFORMANCE ANALYSIS:

Marketing performance analysis shows a strong relationship between marketing investment and revenue generation. Higher spending generally leads to increased revenue; however, the relationship is not always consistent.

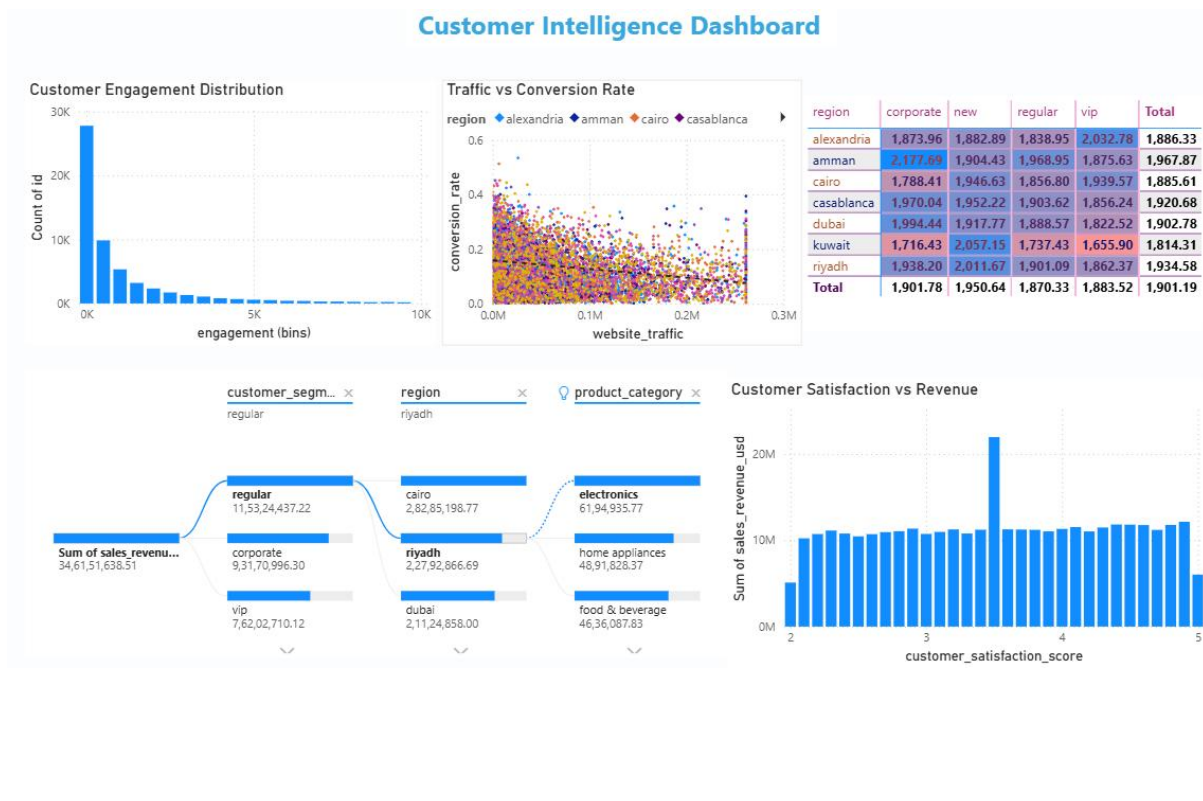
The variation in ROI indicates that some campaigns are highly effective, while others are less efficient. This suggests the need for better optimization of marketing strategies and budget allocation.



3. CUSTOMER BEHAVIOR ANALYSIS:

Customer behavior analysis reveals that engagement levels are highly skewed, with most customers showing low interaction and only a small group contributing significantly.

The relationship between website traffic and conversion rate is weak, indicating that higher traffic does not guarantee better sales performance. Customer satisfaction shows limited influence on revenue, suggesting other factors play a more dominant role.



CONCLUSION

The project successfully demonstrates the complete workflow from raw data preprocessing to automated analysis and visualization. While marketing investment is a key driver of revenue, the findings indicate the need for improved efficiency in campaign performance and customer conversion.

The insights generated from this analysis can support data-driven decision-making, enabling better allocation of resources and more targeted business strategies.